

Moving Average Strategy Effectiveness Across P/E Ratio Regimes: Evidence from Six U.S. Equities (2020–2025)

Reid Fintak

Independent Researcher

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Abstract

This paper investigates whether a stock's price-to-earnings (P/E) ratio regime, defined as high or low relative to a stock-specific historical median, predicts the relative effectiveness of moving average (MA) crossover strategies against a passive buy-and-hold benchmark. Using daily price data and verified GAAP diluted earnings per share for six U.S. equities across five sectors from January 2020 to January 2025, I test four standard MA strategies: price vs. 20-day moving average, 20-day vs. 50-day, 10-day vs. 20-day, and 5-day vs. 20-day. Regime classification uses both median and quartile P/E splits. Statistical significance is assessed using Welch's t-tests; effect size is measured with Cohen's d. Results indicate that the hypothesized pattern, namely that MA strategies perform best relative to buy-and-hold in low P/E environments, holds most consistently and significantly for technology stocks (Apple and Microsoft), with directional but

weaker support for financial services (JPMorgan Chase) and consumer staples (Coca-Cola). The effect is absent for energy (ExxonMobil) and healthcare (Johnson & Johnson). Effect sizes are uniformly small (Cohen's $d \leq 0.14$), consistent with the noise inherent in daily equity returns. These findings suggest P/E regime has sector-dependent, modest predictive value for MA strategy effectiveness.

Keywords: moving averages, technical analysis, P/E ratio, market regimes, equity strategy, buy-and-hold

1. Introduction

The efficient market hypothesis (EMH), introduced by Fama (1970), posits that asset prices fully reflect all available information, implying that no systematic trading strategy can consistently outperform a passive benchmark on a risk-adjusted basis. Technical analysis, the practice of using historical price patterns and indicators to generate trading signals, is theoretically incompatible with strong-form market efficiency. Yet technical indicators such as moving averages remain among the most widely used tools in practice, and a substantial empirical literature documents mixed evidence regarding their effectiveness.

Moving average strategies have received considerable academic attention. The seminal work of Brock, Lakonishok, and LeBaron (1992) demonstrated that simple MA rules generated positive returns on Dow Jones Industrial Average data from 1897 to 1986. Subsequent research showed that these results are sensitive to time period, transaction costs, and data-snooping biases (Sullivan, Timmermann, & White, 1999). More recent work has found that MA strategies retain

value under specific market conditions, particularly in trending environments, while performing poorly in flat or mean-reverting markets.

A separate strand of literature examines fundamental valuation metrics, particularly the price-to-earnings (P/E) ratio, as predictors of equity returns. Campbell and Shiller (1988) demonstrated that high P/E ratios historically predict lower long-run market returns, consistent with mean-reverting valuation dynamics. However, the interaction between P/E regimes and technical strategy effectiveness at the individual-stock level has received limited direct attention.

This paper addresses one specific, underexplored question: does a stock's P/E ratio regime predict whether MA strategies outperform buy-and-hold? The underlying logic is straightforward. In high P/E environments, often associated with sustained price momentum and investor optimism, passive holding may benefit more from continued appreciation than any active switching strategy. In low P/E environments, often associated with market stress, uncertainty, or early recovery, trend signals may provide real timing value by reducing equity exposure during drawdowns and re-entering during recoveries.

I test this hypothesis using six U.S. equities spanning five sectors over a five-year window that includes notable market diversity: the COVID-19 crash and recovery (2020), a sustained bull market (2021), a rate-driven drawdown (2022), and a multi-year recovery (2023–2024). This variation provides the regime diversity needed to identify condition-dependent effects.

My main findings are twofold. The hypothesized pattern holds consistently for technology stocks and directionally for financial and consumer staples stocks, but breaks down for energy and healthcare, a sector-dependent result with a clear economic explanation.

Statistical significance is weak at the daily level, with uniformly small effect sizes, consistent with the efficient-markets prior.

2. Literature Review

2.1 Moving Averages and Technical Analysis

Moving average strategies operate on the premise that prices exhibit persistence, meaning that recent trends carry information about future direction. The most common implementations are: (1) price vs. moving average, generating a long signal when price crosses above the MA; and (2) dual crossover strategies, generating a long signal when a shorter MA crosses above a longer MA. Both are trend-following in nature.

Brock, Lakonishok, and LeBaron (1992) provided early systematic evidence that simple MA rules generated excess returns on DJIA data over a 90-year period. Sullivan, Timmermann, and White (1999) demonstrated that these results are vulnerable to data-snooping bias: when evaluated across the full universe of possible trading rules, the apparent success of select strategies reflects selection rather than genuine predictive power. Zhu and Zhou (2009) showed that MA rules nonetheless retain value as portfolio allocation tools, demonstrating theoretically that MAs function as fast learning methods about underlying return dynamics when predictability is uncertain. Lo, Mamaysky, and Wang (2000) applied systematic pattern recognition methods to a large sample of U.S. stocks from 1962 to 1996 and found that several technical indicators provide incremental information beyond what standard return models capture.

This literature collectively suggests that MA performance is condition-dependent rather than uniformly positive or negative, motivating the use of a fundamental valuation metric as a conditioning variable.

2.2 P/E Ratio and Equity Returns

The price-to-earnings ratio is among the most widely cited equity valuation metrics. Campbell and Shiller (1988) demonstrated that high P/E ratios predict lower long-run market returns, consistent with mean-reverting valuation dynamics. At the individual-stock level, La Porta (1996) showed that investment strategies exploiting overoptimistic analyst growth expectations embedded in high P/E stocks earn superior returns, because expectations about future earnings growth are systematically too extreme. In effect, high P/E stocks underperform over multi-year horizons because their prices embed unrealizable optimism. At shorter horizons, however, momentum tends to dominate valuation signals (Jegadeesh & Titman, 1993), creating a tension between momentum and mean reversion that the present paper exploits directly. In high P/E environments, momentum tends to dominate; in low P/E environments, valuation-driven correction may exert more influence, conditions with different implications for trend-following strategies.

2.3 The Gap Addressed

To my knowledge, the specific interaction between P/E regime and MA strategy effectiveness at the individual-stock level has not been systematically studied. Prior work in the factor-investing literature has examined combined fundamental and technical signals, but has not tested whether a stock's current P/E level predicts when MA strategies outperform passive

holding across multiple sectors. This paper contributes a cross-sector test of that hypothesis with formal statistical evaluation and transparent reporting of effect sizes.

3. Data and Methodology

3.1 Data Collection

Daily OHLCV data for six U.S. equities, Apple (AAPL), Microsoft (MSFT), ExxonMobil (XOM), JPMorgan Chase (JPM), Coca-Cola (KO), and Johnson & Johnson (JNJ), were obtained from Yahoo Finance via the `yfinance` Python library, covering January 1, 2020 to January 1, 2025 (approximately 1,258 trading days per stock). These stocks provide coverage across five sectors: technology (AAPL, MSFT), energy (XOM), financial services (JPM), consumer staples (KO), and healthcare (JNJ).

Quarterly GAAP diluted earnings per share were collected manually from SEC 8-K filings for each company, covering 20–21 quarters per stock. All values were cross-verified against `macrotrends.net` historical EPS records and validated against published annual totals. Full data and source code are available in the accompanying GitHub repository.

3.2 TTM EPS and P/E Ratio Computation

All EPS data use report dates, the dates on which earnings were publicly announced, rather than fiscal quarter-end dates. Using quarter-end dates would assign earnings information to trading days before it was available to market participants, introducing lookahead bias into the regime classification. This distinction is fundamental to the integrity of the analysis.

Trailing twelve-month (TTM) EPS at each report date is the sum of the four most recently reported quarterly GAAP diluted EPS figures. This metric avoids seasonal distortions from individual quarters and is consistent with standard analyst practice. TTM EPS is forward-filled from each report date to all subsequent trading days until the next report. The daily P/E ratio is then:

$$P/E(t) = \text{Close}(t) / \text{TTM_EPS}(\text{report_date})$$

Days where TTM EPS is zero or negative are excluded from P/E analysis, as the ratio is undefined for negative earnings. This primarily affects ExxonMobil due to a large Q4 2020 write-down (-\$4.70 per share), reducing XOM's valid-day count to 733 of 1,258. Two anomalous EPS observations are noted: JNJ Q3 2023 (\$10.21) reflects a one-time talc litigation settlement gain that inflates TTM EPS for approximately one year thereafter; JNJ Q1 2023 (-\$0.03) reflects a one-time charge. Both are retained as reported for GAAP consistency, with their potential distortive effects discussed in the Limitations section.

3.3 Moving Average Strategies

Four standard MA crossover strategies are tested. Each generates a daily signal of +1 (long), -1 (short), or 0 (flat) during the initial MA formation period at the start of the sample:

Strategy 1 (Price vs. MA20): Long when $\text{Close} > \text{SMA}(20)$; short when $\text{Close} < \text{SMA}(20)$.

Strategy 2 (MA20 vs. MA50): Long when $\text{SMA}(20) > \text{SMA}(50)$; short otherwise.

Strategy 3 (MA10 vs. MA20): Long when $\text{SMA}(10) > \text{SMA}(20)$; short otherwise.

Strategy 4 (MA5 vs. MA20): Long when $\text{SMA}(5) > \text{SMA}(20)$; short otherwise.

These four strategies span from fast (MA5/MA20) to slow (MA20/MA50), covering the speed range most commonly studied in the technical analysis literature.

3.4 Backtest Mechanics

Each strategy is simulated over the full sample period starting with \$1,000 in capital. Positions execute at the closing price on the signal day, with a symmetric 0.1% transaction fee applied to each buy and sell. Portfolios are either fully invested in the equity or fully in cash; there are no partial positions or leverage. A buy-and-hold benchmark is computed as \$1,000 invested at the opening price on January 1, 2020 and held throughout, with no fees. Daily portfolio value is cash plus shares multiplied by closing price. Daily returns are percentage changes in portfolio value.

3.5 Regime Classification and Statistical Tests

Trading days are classified into P/E regimes using two methods. Under the primary median split, days with P/E below the stock's full-period sample median are classified as low P/E and days above as high P/E, producing approximately equal group sizes. Under the quartile robustness check, only days in the bottom quartile (P/E at or below the 25th percentile) are classified as low P/E and only days in the top quartile (P/E at or above the 75th percentile) as high P/E; intermediate days are excluded. This tests whether any effect is concentrated at valuation extremes or present across the full distribution.

For each strategy and stock, the daily return edge is the average daily strategy return minus the average daily buy-and-hold return within the same regime. The hypothesis predicts a greater edge, or less negative edge, in low P/E regimes than in high P/E regimes. Statistical significance is assessed using Welch's t-test, which does not assume equal variance between

groups. Cohen's d measures effect size. A result supports the hypothesis ([+]) only when both conditions hold: $p < 0.05$ and the low P/E edge exceeds the high P/E edge.

4. Results

Table 1 summarizes P/E ratio characteristics for each stock over the sample period.

Table 1. P/E Ratio Summary by Stock (January 2020 – January 2025)

Ticker	Sector	P/E Range	Median P/E	Valid Days
AAPL	Technology	20.1 – 42.4	28.8	1,049
MSFT	Technology	22.4 – 39.9	33.8	1,050
XOM	Energy	6.2 – 15.2	9.7	733†
JPM	Financial Services	7.5 – 15.6	9.8	996
KO	Consumer Staples	20.1 – 29.3	24.4	1,054
JNJ	Healthcare	9.0 – 32.4*	21.5	1,061

* JNJ range includes inflation from Q3 2023 one-time litigation settlement. † XOM valid-day count reduced by exclusion of negative-TTM-EPS periods (2020–2021).

4.1 Technology Stocks (AAPL, MSFT)

Apple (AAPL). Under the median split ($p50 = 28.8$), two strategies show statistically significant support for the hypothesis. Price vs. MA20 produces a low P/E edge of +0.0005% against a high P/E edge of -0.0350% ($p = 0.036$, $d = 0.130$). MA10 vs. MA20 produces edges of -0.0520% and -0.0768% respectively ($p = 0.024$, $d = 0.140$). All four strategies show the correct directional pattern under the median split. The quartile robustness check adds one further significant result (MA20 vs. MA50, $p = 0.045$ with correct direction).

Microsoft (MSFT). MSFT produces the most consistent results in the dataset. Under the median split ($p50 = 33.8$), all four strategies are statistically significant ($p = 0.002$ to 0.015) with correct direction. Under the quartile split, three of four remain significant, with p as low as 0.0003. MSFT is the only stock where all four strategies pass both significance and direction criteria under the median split, and the only stock with three or more strategies passing under the quartile split. Cohen’s d reaches 0.140, the largest effect observed in the study.

4.2 Energy (XOM)

Under the median split ($p_{50} = 9.7$), no strategies are statistically significant ($p = 0.165$ to 0.405). Two strategies show the correct direction and two do not, approximately consistent with chance. Under the quartile split, MA20 vs. MA50 and MA10 vs. MA20 become marginally significant with correct direction ($p = 0.043$ and 0.045), suggesting a weak effect may exist at valuation extremes. Given the reduced valid-day count (733) and the absence of significance under the primary split, these results warrant caution.

4.3 Financial Services (JPM)

Under the median split ($p_{50} = 9.8$), all four strategies show correct direction but none reach statistical significance ($p = 0.111$ to 0.677). The quartile split produces a qualitatively different picture: three of four strategies become significant (MA20/MA50: $p = 0.021$; MA10/MA20: $p = 0.002$; MA5/MA20: $p = 0.009$) with correct direction. This pattern, directional at the median but significant only at the extremes, is consistent with a real but modest effect that requires pronounced valuation differences to be detectable.

4.4 Consumer Staples (KO)

KO shows mixed results that weaken under robustness testing. Under the median split ($p_{50} = 24.4$), Price vs. MA20 is significant with correct direction ($p = 0.017$). Three other strategies are significant with the wrong direction, contradicting the hypothesis. Under the quartile split, Price vs. MA20 remains correct-direction significant, but results across other strategies are inconsistent. KO's narrow P/E range (20.1 to 29.3) may limit the meaningfulness of regime splits, as even low P/E days reflect relatively elevated absolute valuations. KO results are treated as inconclusive.

4.5 Healthcare (JNJ)

JNJ produces no statistically significant results under either split method ($p = 0.256$ to 0.904 under the median split). All four strategies show the correct directional pattern, but effect sizes are the smallest in the dataset. The Q3 2023 settlement item introduces artificial P/E elevation for approximately one year, adding noise to the regime classification. JNJ results are treated as inconclusive.

4.6 Cross-Stock Summary

Table 2 summarizes hypothesis support across all six stocks under the median split.

Table 2. Hypothesis Support by Stock and Strategy — Median P/E Split

Ticker	Sector	Pr>MA20	MA20>50	MA10>20	MA5>20
AAPL	Technology	[+]	[~]	[+]	[~]
MSFT	Technology	[+]	[+]	[+]	[+]
XOM	Energy	[~]	[~]	[x]	[x]
JPM	Financial Services	[~]	[~]	[~]	[~]
KO	Consumer Staples	[+]	[x]	[x]	[x]
JNJ	Healthcare	[~]	[~]	[~]	[~]

[+] = $p < 0.05$ and correct direction. [~] = correct direction only. [x] = wrong direction.

Of 24 total tests (6 stocks $\times$ 4 strategies), 7 show [+], 12 show [~], and 5 show [x]. The five [x] results are concentrated in XOM (2 of 4) and KO (3 of 4). No strategy produces a [x] result for AAPL, MSFT, or JPM, indicating sector-level directional consistency even where statistical significance is absent.

5. Discussion

5.1 Why Technology Shows the Strongest Effect

The most consistent support for the hypothesis comes from AAPL and MSFT. High P/E periods for these stocks were concentrated in 2021 and 2023–2024, corresponding to sustained momentum-driven bull markets where passive holding captured strong, uninterrupted gains. MA strategies tend to generate whipsaw losses during brief dips in such environments, exiting and re-entering at higher prices, which erodes returns relative to simply holding.

Low P/E periods for AAPL and MSFT corresponded primarily to late 2020 (prices had not yet fully re-rated following the COVID-19 recovery) and 2022 (the Federal Reserve’s aggressive rate increases drove a sustained drawdown). In these environments, trend signals

provided genuine timing value by reducing equity exposure during the decline and re-entering during the recovery. The median P/E split thus approximately aligns with economically meaningful market conditions, giving the regime classification real interpretive weight.

Technology company earnings are also comparatively stable and predictable. For AAPL and MSFT, EPS grew steadily across the period, meaning TTM P/E primarily reflects investor sentiment about future growth prospects. That signal has economic content for the momentum-versus-mean-reversion dynamic that underpins MA strategy performance.

5.2 Why Energy and Healthcare Break Down

ExxonMobil's price is driven primarily by global oil prices, which are exogenous to the company's P/E ratio. Between 2020 and 2025, XOM's quarterly EPS ranged from -\$4.70 to +\$4.68, driven by pandemic-related write-downs and the 2022 energy price surge. P/E fluctuations for XOM therefore reflect commodity cycles rather than stable equity valuation regimes, removing P/E's predictive content for trend-following performance.

JNJ's inconclusive results stem from two factors. First, the Q3 2023 settlement item contaminates approximately one year of P/E data. Second, JNJ's price exhibited unusually low directional movement throughout the period, providing insufficient variance for the hypothesis test to have adequate statistical power.

5.3 Effect Sizes and Economic Significance

Cohen's d values across all significant results range from 0.025 to 0.140, all classified as small by conventional standards ($d < 0.20$). This is consistent with two established properties of daily financial return data. First, the high day-to-day noise in equity prices attenuates even real

effects at daily frequency. Second, efficient market theory predicts that exploitable predictability will be limited in magnitude. Whether the observed edges are large enough to be actionable after taxes, bid-ask spreads, and execution constraints is outside the scope of this paper.

5.4 Non-Monotonicity of MA Speed

One exploratory finding is worth noting. On AAPL, the fast Price vs. MA20 strategy substantially outperformed the slow MA20 vs. MA50 strategy over the full period (210% vs. 101%), while the ranking reversed on XOM. This non-monotonicity, where optimal MA speed depends on the underlying stock and regime, is consistent with Zhu and Zhou (2009) and reinforces the conclusion that conditioning strategy selection on market context, rather than applying fixed parameters universally, is theoretically motivated.

6. Limitations

Several limitations constrain the generalizability of these findings.

Sample size. Six stocks is a small cross-section. While sector coverage provides breadth, the findings should be considered exploratory and hypothesis-generating rather than confirmatory. A cross-sectional study covering 50 or more equities would substantially strengthen inference.

Time period. The 2020–2025 window is a single, historically unusual period dominated by a pandemic, extreme rate changes, and an AI-driven equity boom. Results may not generalize to other macroeconomic regimes.

Statistical independence. Welch's t-tests treat daily returns as independent observations. Daily equity returns exhibit autocorrelation and volatility clustering, which can cause standard tests to overstate significance. Block-resampled or bootstrapped tests would provide more conservative inference.

Regime threshold. The median P/E split is natural but arbitrary. The quartile robustness check partially addresses this; a continuous or rolling-window regime specification would be more rigorous.

One-time EPS items. JNJ Q3 2023 and XOM Q4 2020 reflect non-recurring accounting events that affect P/E classifications in ways not representative of genuine valuation regimes. Adjusting for these items would improve data quality but introduces subjectivity in classification.

Long-only simulation. The backtest holds cash during -1 signals rather than taking short positions, since most retail investors do not short individual equities. True short positions would produce different results, particularly during the 2022 drawdown.

Single-stock concentration. Each backtest runs on one stock in isolation. Portfolio-level effects, including diversification, correlation, and position sizing, are not captured.

7. Conclusion

This paper examined whether a stock's trailing P/E ratio regime predicts the relative effectiveness of moving average crossover strategies against a passive buy-and-hold benchmark. Using verified GAAP earnings data, lookahead-protected P/E calculations, and Welch's t-tests across six equities from 2020 to 2025, I find directional support for the hypothesis in five of six

stocks under the median split, with statistically significant support concentrated in technology stocks. Microsoft in particular passes both significance and direction criteria under all four strategies at the median split, and under three of four at the quartile split.

The results carry a clear economic interpretation. High P/E periods for technology stocks correspond to sustained, momentum-driven bull markets where passive holding dominates; low P/E periods correspond to drawdowns and early recoveries where trend signals provide meaningful timing value. This pattern breaks down for energy stocks, where commodity-driven earnings volatility confounds P/E as a regime signal, and for healthcare, where a one-time accounting item and low price volatility limit discriminative power.

Effect sizes are uniformly small, consistent with both the efficient-markets prior and the high noise-to-signal ratio of daily return data. The practical takeaway is not that investors should mechanically switch MA strategies based on P/E. Rather, the findings suggest that treating MA strategies as uniformly effective or uniformly ineffective is too coarse. Sector context and valuation regime appear to matter. Future work might extend this analysis to a broader equity cross-section, longer time horizons, and alternative valuation metrics such as the cyclically adjusted P/E (CAPE) or forward earnings multiples.

Data and Code Availability

All analysis code is publicly available at [GITHUB_URL]. The script *ma_pe_analysis.py* contains all steps from data retrieval to statistical output and can be executed without modification after installing the listed dependencies. Daily price data are retrieved programmatically via the yfinance library and require no separate download. Quarterly GAAP diluted EPS figures are hardcoded in the script at their public report dates, sourced from SEC 8-

K filings and cross-verified against macrotrends.net; this design choice prevents lookahead bias that would result from using fiscal quarter-end dates. The paper is also included in the repository.

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